

Installing FFmpeg on Laravel Cloud

Laravel Cloud is out, and I've already been using it for a few projects. Recently, I ran into a limitation: Laravel Cloud doesn't allow installing persistent dependencies like FFmpeg. While they might offer this feature in the future, for now, we need to get creative.

Using Build Commands to Install FFmpeg

Laravel Cloud provides a **Build Commands** feature that lets you execute shell scripts during the build process. We'll leverage this to install FFmpeg every time the app is deployed.

Initial Build Commands

By default, your build commands should look something like this:

```
1 composer install --no-dev
2
3 # npm ci --audit false
4 # npm run build
```

Now, let's modify these steps to install FFmpeg.

Installing FFmpeg

We'll download and install FFmpeg into the project's home directory (\$HOME/bin) so that it's available during runtime.

```
1 # Install & Setup FFmpeg
2
3 # Create a `bin` directory for custom binaries
4 mkdir -p $HOME/bin
5 cd $HOME/bin
```

```
6
7 # Download the FFmpeg binaries
8 curl -L -o ffmpeg.tar.xz https://johnvansickle.com/ffmpeg/releases/ff-
9
10 # Extract the archive
11 tar -xf ffmpeg.tar.xz
12
13 # Rename the extracted directory
14 mv ffmpeg-*-static ffmpeg
15
16 # Grant execution permissions
17 chmod +x $HOME/bin/ffmpeg/ffmpeg
18
19 # Move back to the project root
20 cd $HOME/html
21
22 # Composer installation
23 composer install --no-dev
24
25 # npm ci --audit false
26 # npm run build
```

Using the Correct Binaries

Now that FFmpeg is installed, you can use it in your Laravel application. If you're using the `php-ffmpeg/php-ffmpeg` package, configure it like this:

```
1 $ffmpeg = FFMpeg\FFMpeg::create([
2     'ffmpeg.binaries' => '/var/www/bin/ffmpeg/ffmpeg',
3     'ffprobe.binaries' => '/var/www/bin/ffmpeg/ffprobe',
4 ]);
```

Refactoring the Installation Script

While adding the installation steps directly into your build commands works, it's cleaner and more maintainable to move them into a dedicated script.

Creating a Deployment Script

Let's create a file at `./deploy/ffmpeg.sh` to handle the installation.

```
1 #!/bin/bash
2 set -e # Exit immediately if any command fails
3
4 # Define the target directory
5 BIN_DIR="$HOME/bin"
6
7 # Ensure the bin directory exists
8 mkdir -p "$BIN_DIR"
9
10 # Change to the bin directory
11 (
12     cd "$BIN_DIR"
13
14     # Download FFmpeg
15     echo "Downloading FFmpeg..."
16     curl --silent --show-error --fail -L -o ffmpeg.tar.xz "https://johnvaneider.com/ffmpeg-static"
17
18     # Extract the archive
19     echo "Extracting FFmpeg..."
20     tar -xf ffmpeg.tar.xz
21
22     # Rename the extracted directory
23     mv ffmpeg-* static ffmpeg
24
25     # Set execute permissions
26     chmod +x ffmpeg/ffmpeg
27
28     # Clean up
29     rm -f ffmpeg.tar.xz
30 )
31
32 echo "FFmpeg installation completed successfully!"
```

Updating Build Commands

Now, update your Laravel Cloud Build Commands to use the script:

```
1 # Install & Setup FFmpeg
2 ./deploy/ffmpeg.sh
3
4 # Composer installation
5 composer install --no-dev
6
7 # npm ci --audit false
8 # npm run build
```

Deploying & Verifying

Push your code and trigger a new deployment. If everything works correctly, your build logs should contain something like this:

```
1 00:00:48 Running build commands ————— Finished
2      Downloading FFmpeg...
3      Extracting FFmpeg...
4      FFmpeg installation completed successfully!
5      Setting up Composer...
6      ... Composer logs ...
```

And that's it! You now have FFmpeg installed and running on Laravel Cloud. 🎉